

# Rohit Kota

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Abu Dhabi, UAE | UAE Golden Visa (Valid till 2035) | Open to relocation & travel

## SUMMARY

Computer Vision and Platform Engineer who turns safety requirements into AI systems that hold up in the field. In just over a year on RASED, ADNOC's computer-vision safety platform, helped take 70+ sites live across eight business units, from desert drilling rigs to offshore operations. Owns the work from HSE requirement to validated safety event, working across Fogsphere and e& teams, and has demoed the platform live at GITEX and ADIPEC. Earlier, at Rakuten Symphony, built the garbage-collection system and S3 object APIs for an erasure-coded object store that replaced MinIO, part of more than \$4M in annual licensing savings, and core features of a cloud-native telco automation platform. Four years of hands-on engineering across C, Python, PyTorch, Kubernetes, and PostgreSQL, from distributed systems to AI platforms and cloud-native infrastructure.

## EDUCATION

**College of Engineering, Pune**  
*Bachelor of Technology in Computer Engineering*

Pune, India  
Aug 2018 – Jun 2022

## SKILLS

- **Languages:** Python, C, JavaScript, SQL, Bash; working knowledge of C++ and Go
- **Computer Vision & ML:** PyTorch, TensorFlow/Keras, OpenCV, CNN and transformer (ViT) detectors, object detection and tracking, dataset annotation and augmentation, evaluation metrics (mAP, precision/recall), hyperparameter tuning, model optimization (quantization, TensorRT), multimodal LLMs and vision-language models (VLMs), prompt engineering and fine-tuning, experiment tracking (MLflow)
- **Platforms & DevOps:** Kubernetes (bare-metal and cloud), Docker, Helm, CI/CD (Jenkins, Azure DevOps), GitOps, Longhorn, Veeam Kasten, NGINX, Linux (Ubuntu), PostgreSQL/Patroni, Redis, MQTT, Sanic, Flask, DAG workflow orchestration (Argo Workflows and Kubeflow Pipelines concepts)
- **Cloud & Integration:** Azure (Entra ID, API Management, DevOps, Container Registry), AWS (EKS, S3, IAM), Keycloak, LDAP/SSO, Power BI, SAP HSE, REST APIs and webhooks
- **Tools:** Git, GDB, ASan, K9s, Postman, Wireshark, Jenkins, Selenium

## WORK EXPERIENCE

### Trutech (Fogsphere delivery partner)

*Computer Vision Engineer (Title: Computer Vision Business Analyst), RASED – ADNOC Video Analytics Platform (Fogsphere)*

Abu Dhabi, UAE  
Aug 2025 – Present

Core engineer on RASED, ADNOC's real-time AI safety platform that runs computer-vision models on live CCTV to detect unsafe acts and conditions, deployed across eight ADNOC business units: Drilling, Onshore, Offshore, Gas, Sour Gas, Distribution, Bourouge, and ADNOC HQ. Work across the platform end to end, from requirements with ADNOC HSE teams and site solution design through computer-vision use-case engineering and validation, backend integrations, Kubernetes and edge deployment, and production troubleshooting, serving as the day-to-day technical interface between ADNOC, Fogsphere engineering, and e& (Etisalat) field teams.

### Computer Vision & Solution Engineering

- Own **requirements for new safety use cases** with ADNOC site managers and HQ HSE teams, turning operational risks into detection definitions, acceptance criteria, and alert routing so every camera has a measurable purpose before deployment.
- Design **site-level solutions:** camera coverage per use case, edge sizing by camera count and GPU load, bandwidth plans for event sync and training clips, network and firewall rules, and HA main/backup edge pairs.
- Build **computer-vision use cases** on Fogsphere's no-code AI workflow builder, combining object detection, person tracking, and zone and dwell-time rules to catch PPE violations, red-zone and line-of-fire breaches, man-down, fire and smoke, and vehicle-personnel proximity.
- Fine-tune and evaluate **detection models in PyTorch** on ADNOC site data, building annotation, augmentation, and pre-processing pipelines and tracking experiments in MLflow, raising mAP and recall under site conditions such as dust, glare, and night footage.
- Optimize **models for edge GPU inference** by calibrating confidence thresholds and regions of interest per camera, exporting quantized TensorRT engines, and releasing versions to edge nodes through the model store, cutting false alarms while maintaining recall.

- Prototype **multimodal LLM (vision-language model) agents** that generate plain-language descriptions of detected events and suppress false alarms before escalation, through prompt design and fine-tuning on annotated site clips, benchmarking against the detector-only baseline to raise alert precision.

### Platform, Backend & DevOps

- Engineer **backend integrations and the event pipeline**: ADNOC-specific event schema fields, PostgreSQL reporting views for Power BI, and incident submission to SAP HSE through Azure API Management, so safety events land in ADNOC systems without manual re-entry.
- Integrate **identity and alerting**: Microsoft Entra ID single sign-on with group-to-role mapping via Keycloak, email through the corporate relay, and SMS through Azure API Management, resolving provisioning and access issues for 800+ users.
- Deploy and operate the **central Fogsphere Manager platform** on an 8-node NVIDIA L40 GPU Kubernetes cluster, running Helm and GitOps-driven upgrades with rollback to a 99% availability target across 2,400 cameras and 200 sites.
- Work with Fogsphere's **DevOps and ML teams on platform architecture**, taking part in reviews of microservice scaling, performance, and capacity as camera counts grow, so upgrades and model releases land without interrupting live detection.
- Commission and run **GPU edge inference nodes** at 70+ drilling rigs, offshore facilities, gas plants, and fuel distribution sites, from staging to camera stream integration and main/backup HA pairs, keeping detection running during VSAT and SD-WAN outages with store-and-forward.
- Run the **fleet-wide edge software release cycle**, planning Docker-based bundle rollouts with Fogsphere engineering and scheduling installs with e& field teams in batched windows, keeping remote desert and offshore rigs on current software versions with fewer site visits and no customer disruption.

### Production, Validation & Customer Delivery

- Troubleshoot **production incidents on live rigs** independently, tracing missed or false alerts across camera streams, edge GPU load and connectivity, workflow configuration, and backend services to the root cause, and driving permanent fixes with Fogsphere engineering.
- Resolve **edge hardware and GPU performance issues** with e& field engineers, from GPU throttling and server health to cameras below minimum spec, and tune GPU load and CUDA settings with Fogsphere to keep each node at full camera capacity.
- Validate **releases and fixes against operational requirements**, running UAT with site HSE teams, replaying positive and negative scenarios on live cameras, confirming event generation and alert routing, and signing off versions for production.
- Act as **ADNOC's day-to-day technical counterpart**, running requirement and review sessions with HSE and IT teams, converting defects and change requests into engineering tickets, training site teams and administrators, and managing expectations on live-rig priorities.
- Demonstrated the **Fogsphere platform at GITEX and ADIPEC**, running live computer-vision demos and walking customers, partners, and industry visitors through detection use cases, event handling, and architecture, adjusting depth for technical and executive audiences.

### Rakuten Symphony (Robin.io)

Associate Software Engineer | C, Python, Kubernetes, PostgreSQL

Pune, India

Jun 2022 – Sep 2024

### Object Storage

- Built core subsystems of an in-house **S3-compatible object store for AI-scale workloads** (training datasets, model checkpoints, and artifacts) that replaced MinIO across production Kubernetes clusters, cutting third-party licensing costs by over \$4M a year.
- Packaged the entire store into a **single-command Kubernetes deployment** using Helm charts with automatic rollback and disk-discovery DaemonSets, bringing up three 3-node synchronously replicated Patroni PostgreSQL clusters.
- Designed the **distributed garbage-collection system**, combining a transactional PostgreSQL write-ahead delete queue with at-least-once dispatch and a multithreaded disk scanner that checks chunks against metadata, reclaiming orphaned space even through crashes and restarts.
- Implemented **four S3 object APIs** to AWS spec on librasurecode Reed-Solomon (4+2) striping: zero-copy CopyObject using link(2) hardlinks, HeadObject, ListMultipartUploads, and RFC-7233 ranged GETs over block-aligned O\_DIRECT reads, the byte-range pattern ML data loaders use on sharded datasets.
- Owned the **license management system**, issuing RSA-4096-encrypted keys through a Flask service with Evaluation, Production, and Community tiers, per-cluster capacity limits up to 1 PiB, and expiry checks.
- Developed **cross-site configuration sync** for multi-cluster replication, a REST API that registers remote peer clusters and keeps their configuration consistent in PostgreSQL, surviving database crashes and peer outages through durable work queues and bounded retries.
- Wrote and ran **failure-mode and HA test plans** covering pod, node, and filesystem-full failures, network partitions, and Patroni PostgreSQL failover, verifying end-to-end S3 availability on bare metal with the AWS CLI.

### MDCAP - Multi Data Center Automation Platform (metal-to-service orchestration engine behind Automation Studio)

- Built core features of a **cloud-native automation platform for telco infrastructure fleets** (async Python/Sanic, PostgreSQL, Kubernetes), running declarative DAG workflows for Day-0 to Day-2 provisioning, upgrades, and migrations across bare-metal servers and Kubernetes clusters, executed on containerized worker pods.

- Shrank the platform's **core container image by 90%** (1.78 GB to 180 MB) with a multi-stage build shipping PyInstaller single-file binaries on a minimal Alpine base instead of a full Python runtime and source tree, cutting the attack surface to close a customer security escalation.
- Extended the **declarative workflow DSL and DAG orchestration engine**, the same class of engine as Argo Workflows and Kubeflow Pipelines, with serial and parallel loop unrolling and task grouping, then used it to build a 55-step resumable workflow migrating telco infrastructure from Cisco VIM to Robin's cloud-native platform.
- Cut **fleet dashboard load time at scale** by collapsing per-resource PostgreSQL count queries into one PL/pgSQL aggregation that returns planner estimates instead of exact counts above 10,000 rows, backed by a 5-minute incremental stats snapshot.
- Implemented **least-privilege RBAC** by encoding per-resource-type CRUD bitmaps in JWT claims and enforcing them in API middleware by HTTP verb, creating tiered Admin, Moderator, and Viewer roles across 20+ resource types for 1,000+ clients.
- Developed **secure browser-based access to servers and BMCs** by customizing the Wetty web terminal behind an Nginx WebSocket proxy, with short-lived per-session tokens, no credentials in process arguments, and on-demand ipmitool power control for out-of-band management.
- Added **hybrid-cloud fleet support by onboarding Amazon EKS clusters** as managed resources alongside on-prem Kubernetes, extending the shared resource model with an EKS schema, IAM-based authentication, and lifecycle workflows for cluster operations and network services.

## INTERNSHIP

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### Algoanalytics

Research Intern – Cloud

Pune, India

Jul 2021 – Aug 2021

- Evaluated **cloud AI/ML platforms for hosting AlgoAnalytics' models**, comparing managed training and inference services, GPU compute options, storage, and pricing models across AWS, Azure, and Google Cloud, and recommended deployment targets by cost and workload fit.
- Mapped the **architectures and technology stacks of competing AI products** in finance, healthcare, and retail, from data pipelines and model serving to cloud dependencies and APIs, and turned the findings into a gap analysis and deployment recommendations for the product team.

## ACADEMIC PROJECTS

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- **Xv6 Scheduling Algorithms**: extended the MIT xv6 operating system with new system calls and scheduling algorithms (FCFS, Priority Based Scheduling, Lottery Scheduling, MLFQ), comparing and analyzing their performance.
- **Linux Shell**: developed a command-line interpreter in C demonstrating process control and inter-process communication on UNIX-like systems, with pipes, I/O redirection, shell built-ins, and job control.
- **HTTP Server**: implemented a multithreaded HTTP/1.1-compliant web server in Python supporting GET, POST, PUT, HEAD, and DELETE, cookies, headers, non-persistent connections, concurrent clients, leveled logging, file permissions, server configuration, connection limits, and stop/restart.
- **Fake News Detection**: built a Python classifier over a labeled corpus using NLTK preprocessing (stopword removal, stemming), bag-of-words and TF-IDF features, and Naive Bayes, Logistic Regression, SVM, SGD, and Random Forest models, selecting the best by F1 score and accuracy.
- **Hospital Management System**: developed a MySQL-based system to manage patient records and hospital workflows.
- **Grep**: implemented a multithreaded grep in C that searches multi-gigabyte files faster than the original single-threaded grep.
- **Career Grip**: created a MERN stack web app for managing job applications and recruiting workflows.

## PUBLICATIONS

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### Electronic Health Record Management using Hyperledger Fabric

IEEE ICCUBEA 2022

[ieeexplore.ieee.org/document/10010988](https://ieeexplore.ieee.org/document/10010988)

6th International Conference on Computing, Communication, Control and Automation (ICCUBEA)

First author, Aug 2022

- Designed a **patient-controlled electronic health record system on Hyperledger Fabric**, a permissioned ledger where patients grant and revoke doctors' access and every update is recorded immutably, so records follow the patient across hospitals.
- Showed how the design meets the **security, integrity, confidentiality, and privacy requirements** of sharing sensitive medical data, using certificate-based identities, membership services, and fault-tolerant consensus in place of a trusted central party.